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CN2022s final: Chap 1-Chap6, 7 problems in 4 pages, 100 points

Note:

1. This is an Online exam., open-book, open-everything. But, you should do it alone and cannot get helps from others.
2. Please upload your answer in one file (only .doc (MS word) or .pdf file is allowed). Be sure to have your name in the first page of your upload file.

1. (10%) The network default gateway applying to a host by DHCP is 192.168.5.33/25. Which option is the valid IP address of this host? Choose the right answer(s)(multiple options are possible). And give a short description about why?

- (a). 192.168.5.155
- (b). 192.168.5.47
- (c). 192.168.6.40
- (d). 192.168.5.132
- (e). 192.168.5.114

2. (10%) efficiency of slotted ALOHA protocol

- (a). Suppose N nodes with many frames to send, each transmits in slot with probability p . Explain that the probability of a given node has success to send in a slot is $p(1-p)^{N-1}$. (2%)
- (b). Continue from (a), show that the probability that any node has success to send in a slot is $f(p; N) = Np(1-p)^{N-1}$. (2%)
- (c). Let p^* is the value to maximizes $f(p; N) = Np(1-p)^{N-1}$. Show that $p^* = 1/N$. (hint: find the p s.t. $\partial f(p; N) / \partial p = 0$) (2%)
- (d). Continue from above (c), show that $\lim_{N \rightarrow \infty} f(p^*; N) = e^{-1} = 0.3679$, where

$$p^* = 1/N. \quad (\text{hint: } \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = e^{-1}) \quad (2\%)$$

- (e). Based on above, for un-slotted ALOHA (a.k.a pure ALOHA) protocol, explain that the probability of a given node has success to send in a slot is $p(1-p)^{N-1} (1-p)^{N-1}$. (2%)

3. (10%) Internet check-sum

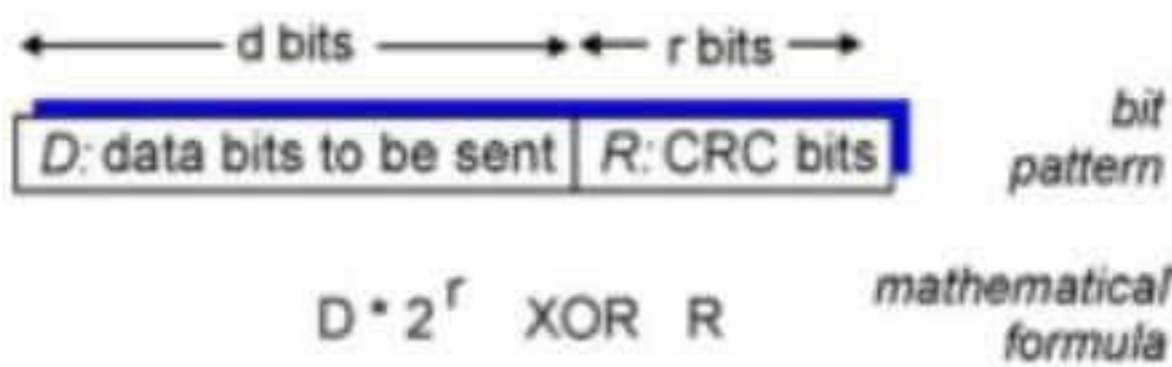
Given a sequent of 12 bytes as (0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0x07, 0x08, 0x09, 0x0A, 0xFF, 0xFF). What is the checksum number of this 12-byte network segment, according to the internet checksum algorithm?

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4. (10%) CRC checksum:

(a). Please explain how CRC checksum works in principle. (hint:

$$R = \text{remainder} \left[\frac{D \cdot 2^r}{G} \right] \quad (4\%)$$



(b). Given, $r=5$, $G=100111$, $D=10111011$, what is R ? (6%)

5. (25%) Suppose one day you bring your notebook to sit in the Computer Network course in our department, you power on your notebook and connected to our department wifi. Then you open a browser with domain name (<https://tronclass.ntou.edu.tw>) to login to the Tronclass course website, and download a course handout. Actually, from the moment you turn on your notebook up to you successfully download the course handout, many of network protocols have been involved between your notebook and different remote nodes. Assume all the network related information cached in your notebook (e.g. arp table, dns resolving table) are empty, and your notebook utilizes dhcp to acquire an IP address.

(a). (16%) From the following network protocols,

http, smtp, pop3, imap, dns, rtp, Skype, dhcp, arp, ospf/rip, please pick four of them which are very likely involved in the above scenario from the time power on up to the moment the handout is download. For each of the four protocols you choose, you should give the following information: 1. The protocol name (what), 2. When it is involved (when), 3. Whom (the remote node) does the notebook communicate with (whom), and 4. Why it should be involved? (why). For example,

1. name: wifi (802.11)
2. when: the time it talks to remote AP
3. whom: the remote AP
4. why: it provides a wireless communication link with remote AP.

(hint: you need both network address and mac address to communicate with other internet devices)

(b) (3%) According to (a), when IP packet will be sent at first time? Explain your answer.

(c) (3%) According to (a), when UDP packet will be sent at first time? Explain

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your answer.

- (d) (3%) According to (a), when TCP packet will be sent at first time? Explain your answer.

6. (30%) Lab:Packet analysis

Figure 1(a) shows an Ethernet frame received, which carries an IP packet as the frame data payload. And, the IP packet embeds an TCP packet. The frame data marked with blue color (that is the byte stream “45 00 00 28 24 92 ...c0 a8 00 64”) is the IP head part, and those followed is TCP packet. Figure 1(b) shows IPv4 packet format. For IPv4, both ver and head length fields take 4 bit, and the flgs field take 3 bits.

(a). Ethernet frame header part (6%)

1. Please check with Ethernet frame format (you may check from book or internet) to get what is the source MAC address carried in this frame packet? (2%)
2. Continue from above, what is the destination MAC address? (2%)
3. Continue from above, how did we know this data frame carries an IP packet in its frame data part? (hint: check the **type** field in the Ethernet header) (2%)

(b). IP packet header part (16%)

1. How should we know this packet is IPv4 but not IPv6 or others? (2%)
2. What is packet header length of this IPv4 packet? (2%)
3. What is total packet length of this IPv4 packet? (2%)
4. What is the TTL(time to live) value of this IPv4 packet? (2%)
5. What is the header checksum value of this IPv4 packet? (2%)
6. How did we know this IP packet carries an TCP datagram in its data part? (hint: check the **upper-layer** field in the IP header) (2%)
7. What is the source IP address carried in this IP packet? (2%)
8. Is this IPv4 packet contains IP header optional part? Why? (2%)

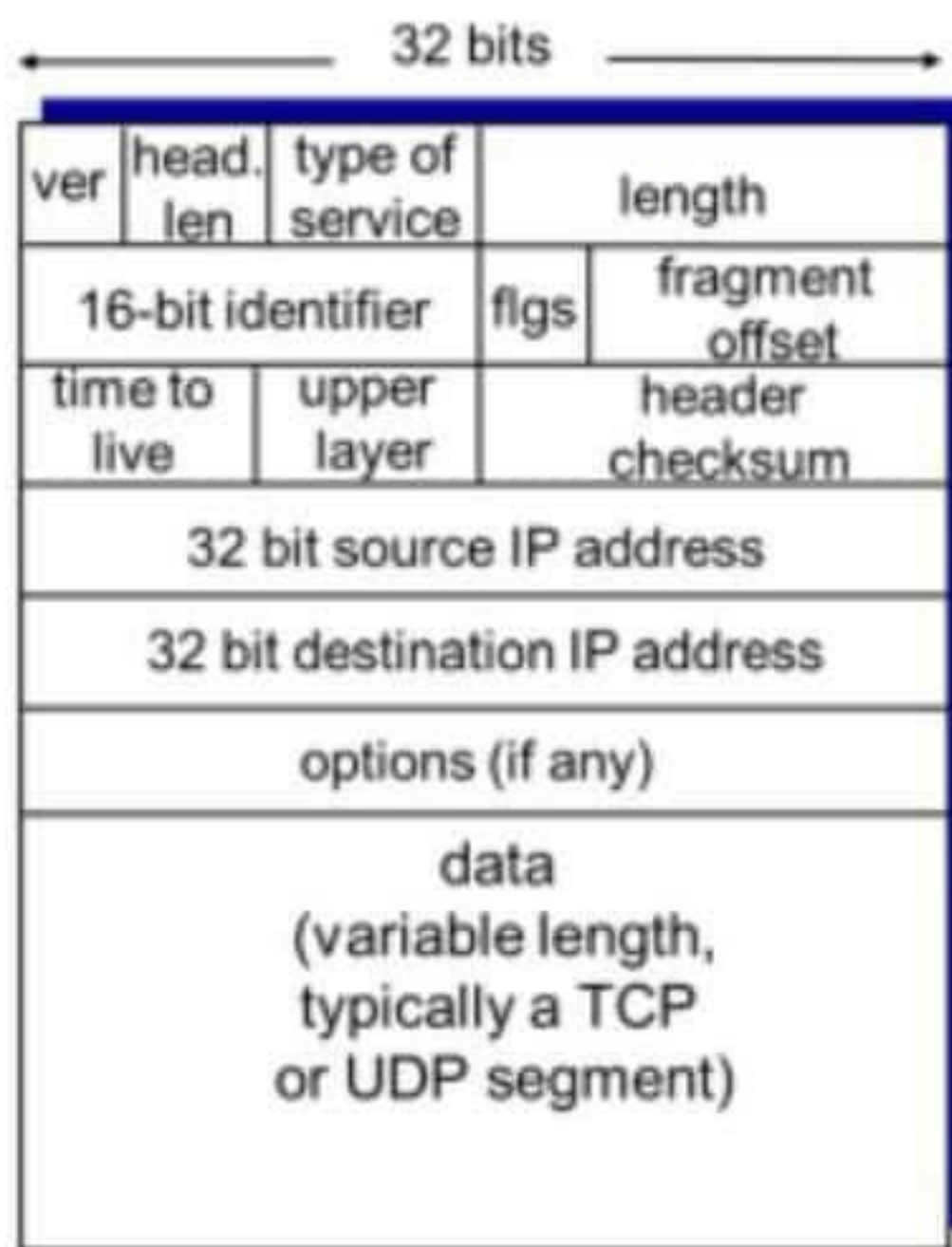
(c). TCP packet part (Please check with TCP packet format from book or internet) (8%):

1. What is the source port and destination port number carried in the TCP header? (hint: check the **TCP** packet format) (2%)
2. What is the TCP header Length of this TCP packet? (2%)
3. What TCP control bits (URG/ACK/PSH/RST/SYN/FIN) are set in this TCP packet? (2%)
4. How many TCP data bytes are carried in this TCP packet?

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0000	f0 2f 74 00 15 7d 18 a6 f7 ab c7 0e 08 00 45 00	./t..}..
0010	00 28 24 92 00 00 79 06 0c 2a ac d9 a3 2e c0 a8	.(\$...y. .*.....
0020	00 64 01 bb f2 9f 1d 85 9b c6 65 c7 45 d7 50 11	.d..... ..e.E.P.
0030	01 09 44 71 00 00 00 00 00 00 00 00	..Dq.....

(a)



(b)

Figure 1. (a) Example of an IPv4 packet encapsulated in an Ethernet frame.

(b) IPv4 packet format. (for Problem 5)

7. (5%) course feedback.

To improve course quality, please give your comments (especially for online course part) if any. Any comments are welcome. No matter what you write, you'll get full score of this question. If you don't know what to write, you can simply write done some short words, like "OK" or "no comment". Thanks!!

End of Questions.

Thanks for having you with us in this course. Happy Summer Break!!